VENTILATION

Job Performance Requirements

NFPA 1010

6.3.11

6.3.12

<u>Note:</u> All objectives in this section are testable.

Introduction

Subject: Tactical Ventilation

Note: All lengths, measurements, and percentages must be verbalized.

1. Tactical ventilation is the planned and systematic removal of heated air, smoke, gases and other airborne contaminants from a structure, and replacing them with cooler and fresh air to meet the incident priorities of life safety, stabilization and property conservation.
2. Confirmation with the Incident Commander (IC) to perform tactical ventilation must be accomplished.
3. Coordination with interior crews and other fire ground activities is vital.
4. The type of tactical ventilation (Horizontal vs. Vertical, and Natural vs. Mechanical) performed, must be the most appropriate for the conditions and situation.
5. Availability of existing openings (Skylights, Ventilator Shafts, Monitors, Hatches) should be utilized whenever possible.
6. If necessary, detection holes should be used to locate the fire. There are two primary types:
    a. Kerf cut (Single cut the width of the saw blade or chain).
    b. Triangle cut

Performance Objective

Subject: Tactical Ventilation

Task: Horizontal (Natural)

Procedure:

1. Confirm with the IC.
2. Confirm the wind direction.
3. Provide an opening on the leeward side at the highest point.
4. Remove any obstructions (blinds, curtains, drapes, screens).
5. Provide an opening on the windward side at the lowest point.
6. Remove any obstructions.

Performance Objective

Subject: Tactical Ventilation

Task: Hydraulic

Procedure:

1. Confirm order with OIC (Officer in Charge).
2. Select the opening to be utilized.
3. Remove any obstructions impeding or around the opening.
4. At the opening, set the nozzle on a wide fog pattern **(Approximately 60 degrees).**
5. **Back away from the opening until the fog stream almost touches the outside edges.**
6. Ensure the fog stream covers 85-90% of the opening.

   Note: The 10-15% area remaining should be located at the top of the opening to allow smoke and heated gases to escape.

7. Continue to flow water through the opening to effect ventilation.

   Note: If necessary, adjust fog pattern and/or distance from opening to effect ventilation.

Performance Objective

Subject: Tactical Ventilation

Task: Positive Pressure

Procedure:

1. Confirm order with command and correspond with the interior crew.

   Note: Coordination is imperative to ensure only one outlet is utilized and NOT have multiple outlets created.

2. The outlet should be opened first and smaller than the size of the inlet.

   Note: Effect is achieved by creating greater air pressure than the surrounding air. If outlet is too large, positive pressure will not be utilized.

3. Remove any obstructions impeding or around the outlet.
4. Position fan approximately **6 feet** from inlet, with blower turned **90 degrees away** from inlet or follow manufacturer's guidelines.
5. If possible, inlet should be on windward side of building.
6. Start the fan, then reposition blower toward the inlet.
7. **Ensure the cone of air is covering 100% of inlet. If necessary, adjust the distance of the fan to obtain full coverage of inlet.**
8. Correspond with interior crew the opening/closing of doors and windows. This will confine/control the movement of smoke and gases.

   Note: Effectiveness can be increased by placing a second fan side by side (same size) or in line (different size) with the first fan. Larger fans should be placed in front of smaller fans closest to the building (between the smaller fan and the building).

Performance Objective

Subject: Tactical Ventilation

Task: Negative Pressure

Procedure:

1. Confirm order with command.
2. Confirm the wind direction.
3. Provide an opening on the leeward side of building at the highest point.
4. Remove any obstructions impeding or around the outlet.
5. Place the "Smoke Ejector" (exhaust fan) in the opening at the highest point.

   Note: Place a salvage cover around the ejector, covering the entire opening to prevent recirculation of smoke and gases into the structure.

6. Provide an opening on the windward side of the building at the lowest point.

Performance Objective

Subject: Tactical Ventilation

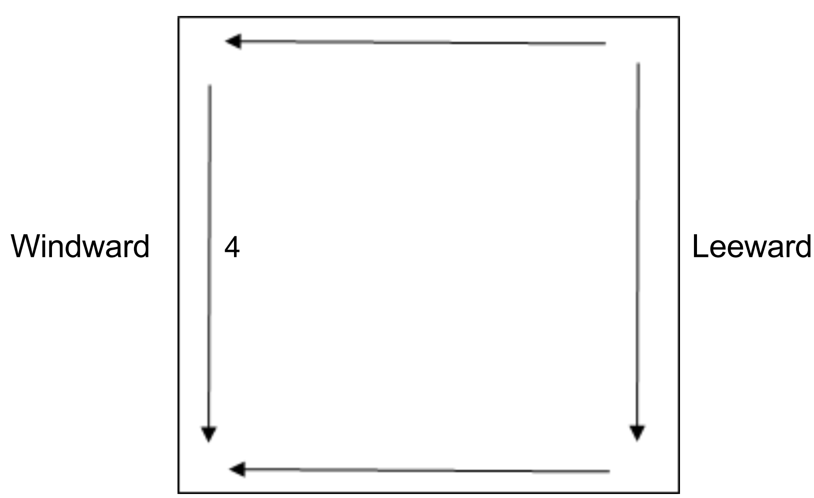
Task: Vertical; Pitched Roof

Procedure:

1. Confirm with the OIC.
2. Confirm wind direction. Work on windward side if possible.
3. Ensure ladder is safe to climb and ascend ladder with tool(s) selected (Pick-Head or Flat-Head Axe, Pike Pole, Roof Ladder, Saw).

   Note: If using a power saw, ensure the saw is operational prior to ascending ladder. Ascend ladder with ONE tool only.

4. Sound the roof for structural integrity prior to stepping on roof.
5. Utilize/place a roof ladder to ensure weight distribution.
6. Select a second means of egress (Escape Route).
7. Locate roof supports (Rafters, Trusses) by sounding at highest point possible and as directly over the fire safely.
8. Remove roof material as necessary.
9. Using a pick-head axe, mark or outline the hole to be cut. Must be at least a 4X4 foot opening.
10. If needed, create a detection hole.
11. Make cuts in the following order due to heated gas and smoke rises:



Note: Depending on instruction, cuts can be done in any order, only if they are performed in a safe manner, and the final cut (#4) is done closest to the ladder.

12. Ensure NOT to cut through the roof supports.
13. Pry up and remove roof decking.
14. Using the blunt end of a pike pole, if necessary, push in the ceiling working from the top of ventilation opening toward bottom.
15. Notify IC when task is demonstrated, remove all tools and descend the ladder.

Performance Objective

Subject: Tactical Ventilation

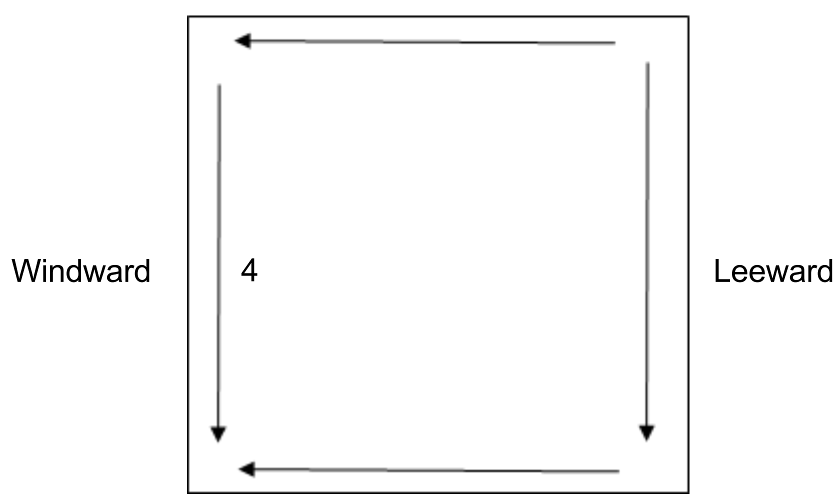
Task: Vertical; Flat Roof

Procedure:

1. Confirm with the OIC.
2. Confirm wind direction. Work on windward side if possible.
3. Ensure ladder is safe to climb and ascend ladder with tool(s) selected (Pick-Head or Flat-Head Axe, Pike Pole, Roof Ladder, Saw).

   Note: If using a Power saw, ensure the saw is operational prior to ascending ladder. Ascend ladder with ONE tool only.

4. Sound the roof for structural integrity prior to stepping on roof.
5. Utilize/place a roof ladder to ensure weight distribution.
6. Select a second means of egress (Escape Route).
7. Locate roof supports (Rafters, Trusses) by sounding as directly over the fire safely.
8. Remove roof material as necessary.
9. Using a pick-head axe, mark or outline the hole to be cut. Must be at least a 4X4 foot opening.
10. If needed, create a detection hole.
11. Make cuts in the following order:



<u>Note:</u> Depending on instruction, cuts can be done in any order, only if they are performed in a safe manner, and the final cut (#4) is done closest to the ladder.

12. Ensure NOT to cut through the roof supports.
13. Pry up and remove roof decking.
14. Using the blunt end of a Pike Pole, if necessary, push in the ceiling working from the farthest side of ventilation opening.
15. Notify IC when task is demonstrated, remove all tools and descend the ladder.